Introduction to Biophysics II: Quantum Biology

Lecture 03
Spin wavefunctions.
Transition between states

- Sin wave functions
- Slater determinant
- Transition between states: time-dependent perturbation theory
- Absorption and emission of light

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Spin wave functions

Spin: two quantum numbers

s = ½ → angular momentum magnitude $S_{magnitude} = \hbar\sqrt{s(s+1)} = \hbar\sqrt{3}/2$

m_s = -s,…,s = -½ , +½ → projection to axis is $S_z = \hbar s = \pm\hbar/2$

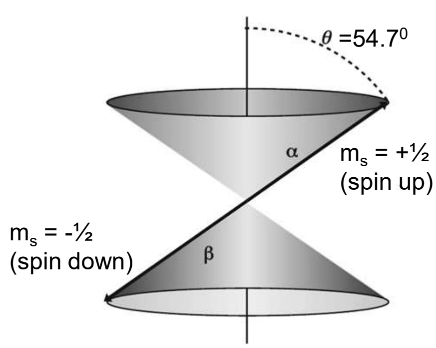
Two spin functions:

$|\alpha\rangle \equiv |\uparrow\rangle$ - spin up

$|\beta\rangle \equiv |\downarrow\rangle$ - spin down

Orthonormality:

$$\langle\alpha|\alpha\rangle = \langle\beta|\beta\rangle = 1$$
$$\langle\alpha|\beta\rangle = \langle\beta|\alpha\rangle = 0$$



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Spin wave functions

Two spin functions: $|\alpha\rangle \equiv |\uparrow\rangle$

$|\beta\rangle \equiv |\downarrow\rangle$

An arbitrary spin state: $|\sigma\rangle = c_1|\uparrow\rangle + c_2|\downarrow\rangle = \sin(\theta)|\uparrow\rangle + \cos(\theta)|\downarrow\rangle$

In matrix form: $|\sigma\rangle = \begin{pmatrix} c_1 \\ c_2 \end{pmatrix}$

$|\alpha\rangle \equiv \begin{pmatrix} 1 \\ 0 \end{pmatrix}$

$|\beta\rangle \equiv \begin{pmatrix} 0 \\ 1 \end{pmatrix}$

Pauli matrices:

$$\sigma_1 = \sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix},$$

$$\sigma_2 = \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix},$$

$$\sigma_3 = \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

Example: expectation value $\langle\sigma|\boldsymbol{\sigma}_z|\sigma\rangle = (c_1 \quad c_2)\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}\begin{pmatrix} c_1 \\ c_2 \end{pmatrix} = c_1^2 - c_2^2$

Combine spatial part (ψ) and spin (σ): $\Psi = \psi(\vec{r})\sigma$

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Two-electron wavefunction

Two electrons: $\Psi_1(1) = \psi_1(1)\sigma_1(1)$

$\Psi_2(2) = \psi_2(2)\sigma_2(2)$

Each has its own spatial and spin wave functions

Two-electron wavefunction – product of the two

However, electrons are indistinguishable, we do not know if 1 is in Ψ_1 or in Ψ_2

How would we build a two-electron wavefunction?

$$\Psi_\pm(1,2) = \frac{1}{\sqrt{2}}\Psi_1(1)\Psi_2(2) \pm \frac{1}{\sqrt{2}}\Psi_1(2)\Psi_2(1)$$

What sign to choose???

Note: $\Psi_+(1,2) = \Psi_+(2,1)$ Symmetric

$\Psi_-(1,2) = -\Psi_-(2,1)$ Antisymmetric

What if we put two electrons in the same state? Should it be *exactly the same?*

Symmetric: yes, it will be the same

Antisymmetric: impossible, unless Ψ=0! No two electrons in exact same state!

Pauli principle: no two electrons can be in exact same state.
Using an antisymmetric combination, we satisfy the Pauli principle.

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Spin wave functions: HOMO

Two electrons on HOMO (highest occupied molecular orbital):

Total S = +1, 0 (parallel or antiparallel)
Total M_S = -S,…,S

For most organic molecules, HOMO contains 2 electrons with antiparallel spins
Since electrons are indistinguishable, the wave function must be a linear combination of both spins (antisymmetric to satisfy Pauli principle)

$$\Psi_a(1,2) = \frac{1}{\sqrt{2}}\Psi_1(1)\Psi_2(2) - \frac{1}{\sqrt{2}}\Psi_1(2)\Psi_2(1)$$

$$\Psi_a(1,2) = \frac{1}{\sqrt{2}}\psi_h(1)\alpha(1)\psi_h(2)\beta(2) - \frac{1}{\sqrt{2}}\psi_h(2)\alpha(2)\psi_h(1)\beta(1)$$

$$\Psi_a = [\psi_h(1)\psi_h(2)]\left[2^{-1/2}\alpha(1)\beta(2) - 2^{-1/2}\alpha(2)\beta(1)\right]$$

Here ψ are the spatial wavefunctions for electron (1) and (2), which depend on the coordinates of the first and second electron, respectively
$\alpha(1)\beta(2)$ means that electron1 has spin up and electron 2 has spin down.

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Spin wave functions: singlet LUMO

HOMO: $\Psi_a = [\psi_h(1)\psi_h(2)]\left[2^{-1/2}\alpha(1)\beta(2) - 2^{-1/2}\alpha(2)\beta(1)\right]$

What happens when we excite one electron to a higher orbital?
Usually, the spin part of the wave function remains the same as for the ground state, so that both S and M_s remain zero.
The spatial part is more complicated:

$$^1\Psi_b = \left[2^{-1/2}\psi_h(1)\psi_l(2) + 2^{-1/2}\psi_h(2)\psi_l(1)\right]\left[2^{-1/2}\alpha(1)\beta(2) - 2^{-1/2}\alpha(2)\beta(1)\right]$$

Here, ψ_h is the spatial function for HOMO, and ψ_b is for LUMO

Why do we have a plus sign for the spatial functions?

Note:
This LUMO is a *singlet* state since spatial functions are symmetric, and there is only one possible combination of spin functions
The ground state is also a singlet

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Spin wave functions: triplet LUMO

HOMO: $\Psi_a = [\psi_h(1)\psi_h(2)]\left[2^{-1/2}\alpha(1)\beta(2) - 2^{-1/2}\alpha(2)\beta(1)\right]$

LUMO: ${}^1\Psi_b = \left[2^{-1/2}\psi_h(1)\psi_l(2) + 2^{-1/2}\psi_h(2)\psi_l(1)\right]\left[2^{-1/2}\alpha(1)\beta(2) - 2^{-1/2}\alpha(2)\beta(1)\right]$

What happens when we choose an antisymmetric combination of the spatial wave functions for LUMO?

$$ {}^3\Psi_b^{+1} = \left[2^{-1/2}\psi_h(1)\psi_l(2) - 2^{-1/2}\psi_h(2)\psi_l(1)\right]\ [\alpha(1)\alpha(2)] $$

$$ {}^3\Psi_b^{0} = \left[2^{-1/2}\psi_h(1)\psi_l(2) - 2^{-1/2}\psi_h(2)\psi_l(1)\right]\left[2^{-1/2}\alpha(1)\beta(2) + 2^{-1/2}\alpha(2)\beta(1)\right] $$

$$ {}^3\Psi_b^{-1} = \left[2^{-1/2}\psi_h(1)\psi_l(2) - 2^{-1/2}\psi_h(2)\psi_l(1)\right]\ [\beta(1)\beta(2)] $$

Triplet state: three possibilities for M_s = -1,0,+1, i.e., S=1

Note: all wave functions must be antisymmetric, so S=0 will not work

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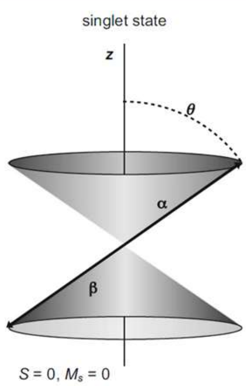
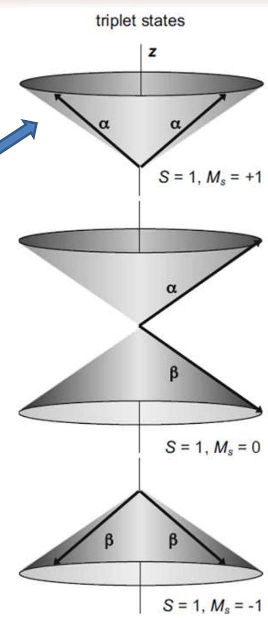
Spin wave functions



$$ S_{magnitude} = \hbar\sqrt{S(S+1)} $$
$$ S_z = \hbar s = \pm\hbar/2 $$

Total spin: on a cone at 45⁰ to z
|S| = $\hbar\sqrt{2}$
$|S_z|$= $\hbar$
S_x – undetermined!



The vectors are drawn to satisfy the quantization of the spin and the z-component of the spin simultaneously for both the individual electrons and the combined system.

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Spin wave functions: energies

Singlet: $$^{1}\Psi_b = \left[2^{-1/2}\psi_h(1)\psi_l(2) + 2^{-1/2}\psi_h(2)\psi_l(1)\right]\left[2^{-1/2}\alpha(1)\beta(2) - 2^{-1/2}\alpha(2)\beta(1)\right]$$

Triplet: $$\left\{\begin{aligned} ^{3}\Psi_b^{+1} &= \left[2^{-1/2}\psi_h(1)\psi_l(2) - 2^{-1/2}\psi_h(2)\psi_l(1)\right]\ [\alpha(1)\alpha(2)] \\ ^{3}\Psi_b^{0} &= \left[2^{-1/2}\psi_h(1)\psi_l(2) - 2^{-1/2}\psi_h(2)\psi_l(1)\right]\left[2^{-1/2}\alpha(1)\beta(2) + 2^{-1/2}\alpha(2)\beta(1)\right] \\ ^{3}\Psi_b^{-1} &= \left[2^{-1/2}\psi_h(1)\psi_l(2) - 2^{-1/2}\psi_h(2)\psi_l(1)\right]\ [\beta(1)\beta(2)] \end{aligned}\right.$$

Singlet-triplet splitting:

- Triplet state $^{3}\Psi_b^{0}$ has *lower* energy than singlet state $^{1}\Psi_b$.

This is a consequence of different *spatial* wavefunctions → the motion of two electrons is correlated in a way that tends to keep electrons further away from each other in triplet, decreasing repulsive interaction *(Hundt's rule).*
The difference is called *singlet-triplet splitting* and is equal to $2K_{hl}$, where

$$K_{hl} = \left\langle \psi_l(1)\psi_h(2) \left| \frac{e^2}{r_{12}} \right| \psi_h(1)\psi_l(2) \right\rangle.$$ ← *Exchange integral (>0)*

Special case: when two orbitals are equivalent in energy and can be transformed into each other by a symmetry operation (such as rotation) → triplet can be lower in energy, like in O_2 (triplet is ground state, singlet is excited).

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Spin wave functions: many electrons

Wave functions for more than two electrons can be approximated as *linear combinations* of all the allowable products of electronic and spin wavefunctions for the individual electrons.

Combinations for N electrons that *automatically* satisfy the Pauli exclusion principle can be written conveniently as a ***Slater determinant*** (1929):

$$\Psi = (1/N!)^{1/2}\begin{vmatrix} \psi_a(1)\alpha(1) & \psi_a(2)\alpha(2) & \psi_a(3)\alpha(3) & \cdots & \psi_a(N)\alpha(N) \\ \psi_a(1)\beta(1) & \psi_a(2)\beta(2) & \psi_a(3)\beta(3) & \cdots & \psi_a(N)\beta(N) \\ \psi_b(1)\alpha(1) & \psi_b(2)\alpha(2) & \psi_b(3)\alpha(3) & \cdots & \psi_b(N)\alpha(N) \\ \psi_b(1)\beta(1) & \psi_b(2)\beta(2) & \psi_b(3)\beta(3) & \cdots & \psi_b(N)\beta(N) \\ \vdots & \vdots & \vdots & & \vdots \\ \psi_N(1)\beta(1) & \psi_N(2)\beta(2) & \psi_N(3)\beta(3) & \cdots & \psi_N(N)\beta(N) \end{vmatrix}$$

John Clarke Slater
1900-1976

where the ψ_i are the individual one-electron spatial wavefunctions

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Spin wave functions: many electrons

Check with 2 electrons:

$$\Psi = (1/N!)^{1/2} \begin{vmatrix} \psi_a(1)\alpha(1) & \psi_a(2)\alpha(2) & \psi_a(3)\alpha(3) & \cdots & \psi_a(N)\alpha(N) \\ \psi_a(1)\beta(1) & \psi_a(2)\beta(2) & \psi_a(3)\beta(3) & \cdots & \psi_a(N)\beta(N) \\ \psi_b(1)\alpha(1) & \psi_b(2)\alpha(2) & \psi_b(3)\alpha(3) & \cdots & \psi_b(N)\alpha(N) \\ \psi_b(1)\beta(1) & \psi_b(2)\beta(2) & \psi_b(3)\beta(3) & \cdots & \psi_b(N)\beta(N) \\ \vdots & \vdots & \vdots & & \vdots \\ \psi_N(1)\beta(1) & \psi_N(2)\beta(2) & \psi_N(3)\beta(3) & \cdots & \psi_N(N)\beta(N) \end{vmatrix}.$$

$$\Psi = \frac{1}{\sqrt{2}} \begin{vmatrix} \psi_a(1)\alpha(1) & \psi_a(2)\alpha(2) \\ \psi_a(1)\beta(1) & \psi_a(2)\beta(2) \end{vmatrix}$$

$$\Psi = \frac{1}{\sqrt{2}}\left[\psi_a(1)\alpha(1)\psi_a(2)\beta(2) - \psi_a(2)\alpha(2)\psi_a(1)\beta(1)\right]$$

Compare with the previous expression for HOMO:

$$\Psi_a = [\psi_h(1)\psi_h(2)]\ \left[2^{-1/2}\alpha(1)\beta(2) - 2^{-1/2}\alpha(2)\beta(1)\right]$$

$$\Psi_a = \frac{1}{\sqrt{2}} \begin{vmatrix} \psi_h(1)\alpha(1) & \psi_h(2)\alpha(2) \\ \psi_h(1)\beta(1) & \psi_h(2)\beta(2) \end{vmatrix}$$

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Spin wave functions: many electrons

Too much to write! →

$$\Psi = (1/N!)^{1/2} \begin{vmatrix} \psi_a(1)\alpha(1) & \psi_a(2)\alpha(2) & \psi_a(3)\alpha(3) & \cdots & \psi_a(N)\alpha(N) \\ \psi_a(1)\beta(1) & \psi_a(2)\beta(2) & \psi_a(3)\beta(3) & \cdots & \psi_a(N)\beta(N) \\ \psi_b(1)\alpha(1) & \psi_b(2)\alpha(2) & \psi_b(3)\alpha(3) & \cdots & \psi_b(N)\alpha(N) \\ \psi_b(1)\beta(1) & \psi_b(2)\beta(2) & \psi_b(3)\beta(3) & \cdots & \psi_b(N)\beta(N) \\ \vdots & \vdots & \vdots & & \vdots \\ \psi_N(1)\beta(1) & \psi_N(2)\beta(2) & \psi_N(3)\beta(3) & \cdots & \psi_N(N)\beta(N) \end{vmatrix}.$$

For brevity:
- 1/N! normalization factor is often omitted
- Only diagonal terms are listed

$$\psi_j(k)\alpha(k) \rightarrow \psi_j(k)$$
$$\psi_j(k)\beta(k) \rightarrow \overline{\psi}_j(k).$$

- Indices for electrons are omitted

Resulting in: $\boxed{\Psi = \left|\psi_a\overline{\psi_a}\psi_b\overline{\psi_b}\cdots\overline{\psi_N}\right|.}$ (means the same matrix!)

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Spin wave functions: many electrons

Too much to write! → $\Psi = (1/N!)^{1/2} \begin{vmatrix} \psi_a(1)\alpha(1) & \psi_a(2)\alpha(2) & \psi_a(3)\alpha(3) & \cdots & \psi_a(N)\alpha(N) \\ \psi_a(1)\beta(1) & \psi_a(2)\beta(2) & \psi_a(3)\beta(3) & \cdots & \psi_a(N)\beta(N) \\ \psi_b(1)\alpha(1) & \psi_b(2)\alpha(2) & \psi_b(3)\alpha(3) & \cdots & \psi_b(N)\alpha(N) \\ \psi_b(1)\beta(1) & \psi_b(2)\beta(2) & \psi_b(3)\beta(3) & \cdots & \psi_b(N)\beta(N) \\ \vdots & \vdots & \vdots & & \vdots \\ \psi_N(1)\beta(1) & \psi_N(2)\beta(2) & \psi_N(3)\beta(3) & \cdots & \psi_N(N)\beta(N) \end{vmatrix}$

$$\boxed{\Psi = \left|\psi_a\overline{\psi_a}\psi_b\overline{\psi_b}\cdots\overline{\psi_N}\right|}$$

Our previous 2-e orbitals:

HOMO: $\Psi_a = \left|\psi_h\overline{\psi_h}\right|,$

LUMO: $^1\Psi_b = 2^{-1/2}\left(\left|\psi_h\overline{\psi_l}\right| + \left|\psi_l\overline{\psi_h}\right|\right)$ $\quad {}^1\Psi_b = \left[2^{-1/2}\psi_h(1)\psi_l(2) + 2^{-1/2}\psi_h(2)\psi_l(1)\right]\left[2^{-1/2}\alpha(1)\beta(2) - 2^{-1/2}\alpha(2)\beta(1)\right]$

$^3\Psi_b^{+1} = 2^{-1/2}\left(\left|\psi_h\psi_l\right| - \left|\psi_l\psi_h\right|\right).$ $\quad {}^3\Psi_b^{+1} = \left[2^{-1/2}\psi_h(1)\psi_l(2) - 2^{-1/2}\psi_h(2)\psi_l(1)\right]\ [\alpha(1)\alpha(2)]$

$^3\Psi_b^{0} = 2^{-1/2}\left(\left|\psi_h\overline{\psi_l}\right| - \left|\psi_l\overline{\psi_h}\right|\right)$ $\quad {}^3\Psi_b^{0} = \left[2^{-1/2}\psi_h(1)\psi_l(2) - 2^{-1/2}\psi_h(2)\psi_l(1)\right]\left[2^{-1/2}\alpha(1)\beta(2) + 2^{-1/2}\alpha(2)\beta(1)\right]$

$^3\Psi_b^{-1} = 2^{-1/2}\left(\left|\overline{\psi_h}\ \overline{\psi_l}\right| - \left|\overline{\psi_l}\ \overline{\psi_h}\right|\right)$ $\quad {}^3\Psi_b^{-1} = \left[2^{-1/2}\psi_h(1)\psi_l(2) - 2^{-1/2}\psi_h(2)\psi_l(1)\right]\ [\beta(1)\beta(2)]$

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Protons and other particles

Protons also have spin s=1/2 and m_s=-1/2, +1/2

Wave functions of systems containing multiple protons, or any other particles with half-integer spin (fermions), also resemble electronic wave functions in being antisymmetric for the interchange of any two identical particles.

Wave functions for systems containing multiple particles with *integer* or zero *spins* (**bosons**), by contrast, must be *symmetric* for the interchange of two identical particles. Example: deuteron (proton+neutron nucleus), S=1.

Bosons: interchanging bosons leaves their combined wave function unchanged, any number of bosons can occupy the same wave function!

Fermions and bosons obey different statistics that become increasingly distinct at low temperatures
(See book to recall Botzmann, Fermi-Dirac and Bose-Einstein statistics)

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Transitions between states: Time-dependent perturbation theory

$$\hat{H}\psi^b(\vec{r}) = E^b\psi^b(\vec{r})$$ ← Solutions to SE describe *stationary states*

SE may have a set of *N* solutions, (ψ_n^b,E_n^b), *n=1…N*. Wave functions are orthonormal and complete, i.e., *any* linear combination of ψ_n is a solution, and *any* possible wave function can be represented as a linear combination of ψ_n^b:

$$\psi_k = \sum_{n=1}^{N} C_n^k \psi_n^b$$

The average/expectation energy is given by the weighted average:

$$E_k = \langle \psi_k | \hat{H} | \psi_k \rangle = \sum_{n=1}^{N} \left|C_n^k\right|^2 \langle \psi_n | \hat{H} | \psi_n \rangle = \sum_{n=1}^{N} \left|C_n^k\right|^2 E_n^b$$

However, when measuring energy, we always get one of the energies for the stationary states with a probability $\left|C_n^k\right|^2$

When an atom/molecule is interacting with a bath (protein environment), it ends up in a state where these probabilities represent the Fermi-Dirac distribution, or at room temperature, ~Boltzmann distribution. Electronic states are generally separated by energy >kT, it will be the lowest single energy (ground) state.

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Transition between states

Let's assume that an electron is placed into one *stationary* state described by Hamiltonian $\mathbf{H=H_0}$.

If there are no interactions with the surroundings (bath), it will stay there forever, as long as Hamiltonian **H** does not change with time, because the energy of the system cannot change.

Suppose **H** changes with time → for example, due to interaction with an electric field or with another molecule.

→ This will perturb the original Hamiltonian, and the state will no longer be stationary; the original solutions to SE are not valid.

Perturbation theory:

Assume that <u>*perturbation*</u> *to the original Hamiltonian* $\mathbf{H_0}$ *is small:*

$$H = H^0 + H'(t).$$

Note1: for technical reasons we will use different hats on top of operators or use bold. But these are the same quantities…

Note2: We will follow Griffiths's Chapter 9 solution since Parson's text is too sloppy

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Transition between states

Small Perturbation" $H = H^0 + H'(t).$

Since perturbation *H'(t)* is small, we will approximate the solution to the new perturbed Hamiltonian as a superposition of solutions to **H^0:**

$$\Psi(t) = c_a \Psi_a(t) + c_b \Psi_b(t) \quad \leftarrow \text{Assume 2-level system} \quad \begin{cases} H^0 \Psi_a = i\hbar \dfrac{\partial \Psi_a}{\partial t} \\ H^0 \Psi_b = i\hbar \dfrac{\partial \Psi_b}{\partial t} \end{cases}$$

Separate variables:

$$\Psi(t) = c_a(t)\psi_a e^{-iE_a t/\hbar} + c_b(t)\psi_b e^{-iE_b t/\hbar}$$

Where c_a and c_b depend on time. Note that $|c_a|^2$ and $|c_b|^2$ represent the extent of Ψ that resembles the original stationary wavefunctions Ψ_a and Ψ_b.

Assume: the molecule is in state Ψ_a before perturbation is introduced, $c_a(0)=1$. How rapidly will the wavefunction Ψ begin to resemble basic state Ψ_b, i.e. $c_b>1$

Need to solve TDSE: $\hat{H}\Psi = i\hbar \dfrac{\partial \Psi}{\partial t}$

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Transition between states

Hamiltonian: $H = H^0 + H'(t).$

Seek solutions: $\Psi(t) = c_a \Psi_a(t) + c_b \Psi_b(t) = c_a(t)\psi_a e^{-iE_a t/\hbar} + c_b(t)\psi_b e^{-iE_b t/\hbar}$

Solve SE: $H\Psi = i\hbar \dfrac{\partial \Psi}{\partial t}$

$$\cancel{c_a[H^0\psi_a]e^{-iE_a t/\hbar}} + \cancel{c_b[H^0\psi_b]e^{-iE_b t/\hbar}} + c_a[H'\psi_a]e^{-iE_a t/\hbar} + c_b[H'\psi_b]e^{-iE_b t/\hbar}$$
$$= i\hbar\Big[\dot{c}_a\psi_a e^{-iE_a t/\hbar} + \dot{c}_b\psi_b e^{-iE_b t/\hbar}$$
$$+ \cancel{c_a\psi_a\left(-\frac{iE_a}{\hbar}\right)e^{-iE_a t/\hbar}} + \cancel{c_b\psi_b\left(-\frac{iE_b}{\hbar}\right)e^{-iE_b t/\hbar}}\Big]$$

$H^0\psi_a = E_a\psi_a$

$$c_a[H'\psi_a]e^{-iE_a t/\hbar} + c_b[H'\psi_b]e^{-iE_b t/\hbar} = i\hbar\left[\dot{c}_a\psi_a e^{-iE_a t/\hbar} + \dot{c}_b\psi_b e^{-iE_b t/\hbar}\right]$$

$<\psi_a|\times$ Need to isolate

$$c_a\langle\psi_a|H'|\psi_a\rangle e^{-iE_a t/\hbar} + c_b\langle\psi_a|H'|\psi_b\rangle e^{-iE_b t/\hbar} = i\hbar\dot{c}_a e^{-iE_a t/\hbar}$$

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Transition between states

Have:

$$c_a\langle\psi_a|H'|\psi_a\rangle e^{-iE_at/\hbar}+c_b\langle\psi_a|H'|\psi_b\rangle e^{-iE_bt/\hbar}=i\hbar\dot{c}_a e^{-iE_at/\hbar}$$

Define: $H'_{ij}\equiv\langle\psi_i|H'|\psi_j\rangle$ Hermiticity: $H'_{ji} = (H'_{ij})^*$

$$\boxed{\dot{c}_a=-\frac{i}{\hbar}\left[c_aH'_{aa}+c_bH'_{ab}e^{-i(E_b-E_a)t/\hbar}\right]}$$

Can isolate c_b derivative in a similar way: $(<\psi_b|\times)$

$$\boxed{\dot{c}_b=-\frac{i}{\hbar}\left[c_bH'_{bb}+c_aH'_{ba}e^{i(E_b-E_a)t/\hbar}\right]}$$

Typically: $H'_{aa}=H'_{bb}=0.$

And we have: $\boxed{\dot{c}_a=-\frac{i}{\hbar}H'_{ab}e^{-i\omega_0t}c_b,\quad \dot{c}_b=-\frac{i}{\hbar}H'_{ba}e^{i\omega_0t}c_a}$

$$\omega_0\equiv\frac{E_b-E_a}{\hbar}$$

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Transition between states

$\boxed{\dot{c}_a=-\frac{i}{\hbar}H'_{ab}e^{-i\omega_0t}c_b,\quad \dot{c}_b=-\frac{i}{\hbar}H'_{ba}e^{i\omega_0t}c_a}$ $\quad \omega_0\equiv\frac{E_b-E_a}{\hbar}$

Initial condition: $c_a(0)=1$, $c_b(0)=0$

Solve:

0th order: $c_a^{(0)}(t)=1,\quad c_b^{(0)}(t)=0$

1st order: insert 0th order solution into right side

$$\frac{dc_a^{(1)}}{dt}=0\Rightarrow c_a^{(1)}(t)=1$$

$$\frac{dc_b^{(1)}}{dt}=-\frac{i}{\hbar}H'_{ba}e^{i\omega_0t}\Rightarrow c_b^{(1)}=-\frac{i}{\hbar}\int_0^t H'_{ba}(t')e^{i\omega_0t'}\,dt'$$

Note:
$|c_a|^2+|c_b|^2\neq 1$
However, it is =1 to first order of H'

2nd order: insert 1st order solution into right side:

$$\frac{dc_a^{(2)}}{dt}=-\frac{i}{\hbar}H'_{ab}e^{-i\omega_0t}\left(-\frac{i}{\hbar}\right)\int_0^t H'_{ba}(t')e^{i\omega_0t'}\,dt'\Rightarrow$$

$$c_a^{(2)}(t)=1-\frac{1}{\hbar^2}\int_0^t H'_{ab}(t')e^{-i\omega_0t'}\left[\int_0^{t'}H'_{ba}(t'')e^{i\omega_0t''}\,dt''\right]dt'$$

$$c_b^{(2)}(t)=c_b^{(1)}(t)$$

Note:
$|c_a|^2+|c_b|^2\neq 1$
However, it is =1 to second order of H'

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Transition between states

Analyze the first order solution:

$$c_b^{(1)} = -\frac{i}{\hbar}\int_0^t H'_{ba}(t')e^{i\omega_0 t'}\,dt'$$

Turn off perturbation after a short time τ
What happens to the system? In what state will it be?

$|c_b(\tau)|^2$ = a statistical probability that the system evolved into state Ψ_b.

Special case:
probability increases linearly with time for small values of τ:

$$|C_b(\tau)|^2 = \kappa\tau$$

Where κ is constant, κ is called a ***rate constant*** for the transition

Transition between states

Analyze the first order solution:

$$c_b^{(1)} = -\frac{i}{\hbar}\int_0^t H'_{ba}(t')e^{i\omega_0 t'}\,dt' \qquad \omega_0 \equiv \frac{E_b - E_a}{\hbar}$$

- Need to specify **H'** to evaluate $C_b(t)$ for specific cases, such as: electromagnetic field, bringing two molecules together, interaction with the surroundings

Things to note:
- If the energies of two states are very different, the *exp*() term oscillates rapidly
- If H'_{ba} is constant, c_b increases linearly only if energies $E_b \approx E_a$ (this is the QM analog of the classical energy conservation principle)
- For absorption of light – the energy difference must match the energy of a photon

Special notes on matrix element $H'_{ba} \equiv \left\langle \psi_b \middle| \widetilde{H}' \middle| \psi_a \right\rangle$

- *an off-diagonal* matrix element of the *time-dependent* perturbation to Hamiltonian
- Perturbation can drive transition between stationary states Ψ_a and Ψ_b, which are not stationary in the presence of **H'**, but not transitions between stationary states of **H_0+H'**.

Transition between states

$$c_b^{(1)} = -\frac{i}{\hbar}\int_0^t H'_{ba}(t')e^{i\omega_0 t'}\,dt' \qquad \omega_0 \equiv \frac{E_b - E_a}{\hbar}$$

Harmonic interaction: $H'(\mathbf{r}, t) = V(\mathbf{r})\cos(\omega t)$

$$H'_{ab} \equiv \langle \psi_a | H' | \psi_b \rangle = V_{ab}\cos(\omega t)$$

$$V_{ab} \equiv \langle \psi_a | V | \psi_b \rangle$$

Work with 1st order only:

$$c_b(t) \cong -\frac{i}{\hbar}V_{ba}\int_0^t \cos(\omega t')e^{i\omega_0 t'}\,dt' = -\frac{i\,V_{ba}}{2\hbar}\int_0^t \left[e^{i(\omega_0+\omega)t'} + e^{i(\omega_0-\omega)t'}\right]dt'$$

$$c_b(t) \cong -\frac{V_{ba}}{2\hbar}\left[\frac{e^{i(\omega_0+\omega)t} - 1}{\omega_0 + \omega} + \frac{e^{i(\omega_0-\omega)t} - 1}{\omega_0 - \omega}\right]$$

Recall: $\cos x = \frac{e^{ix} + e^{-ix}}{2}$

Restrict case to $\omega \approx \omega_0$: $\omega_0 + \omega \gg |\omega_0 - \omega|$

$$c_b(t) \cong -\frac{V_{ba}}{2\hbar}\frac{e^{i(\omega_0-\omega)t/2}}{\omega_0 - \omega}\left[e^{i(\omega_0-\omega)t/2} - e^{-i(\omega_0-\omega)t/2}\right]$$

$$= -i\frac{V_{ba}}{\hbar}\frac{\sin[(\omega_0 - \omega)t/2]}{\omega_0 - \omega}e^{i(\omega_0-\omega)t/2}$$

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Transition between states

Sinusoidal interaction: $H'_{ab} = V_{ab}\cos(\omega t) \quad V_{ab} \equiv \langle \psi_a | V | \psi_b \rangle$

$$c_b(t) \cong -i\frac{V_{ba}}{\hbar}\frac{\sin[(\omega_0 - \omega)t/2]}{\omega_0 - \omega}e^{i(\omega_0-\omega)t/2}$$

Transition probability:

$$\boxed{P_{a\to b}(t) = |c_b(t)|^2 \cong \frac{|V_{ab}|^2}{\hbar^2}\frac{\sin^2[(\omega_0 - \omega)t/2]}{(\omega_0 - \omega)^2}}$$

The probability that particle that was initially in state ψ_a ends up in state ψ_b at time *t*.

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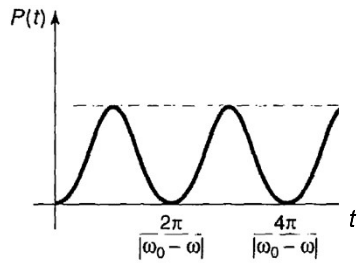
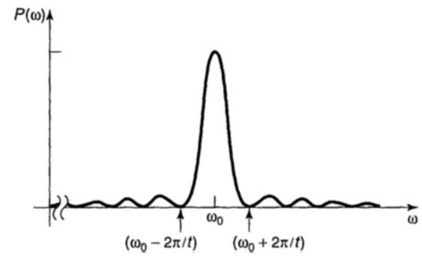
Transition between states

$$P_{a\to b}(t) = |c_b(t)|^2 \cong \frac{|V_{ab}|^2}{\hbar^2}\frac{\sin^2[(\omega_0-\omega)t/2]}{(\omega_0-\omega)^2}.$$





The probability oscillates.
Discuss:
- short perturbation
- long perturbation
- How to get system from a to b most efficiently?

Maximizes at $\omega=\omega_0$
Note: sin(x)/x=1 when x→0

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